PAPER_2135_TILT_SATURATION_1_12_IN_59_OF_116_DISPATCH_OBSERVABLES_MAJORITY_OF_CATALOG_TILT_PRODUCT_LAW_COMPLETE_34_0_PLUS_5_24_FAMILY_30_OBSERVABLES_UQFF_LANDMARK

Daniel T. Murphy

PAPER_2135 — Tilt Saturation: the 1/12 Constant in 59 of 116 Dispatch Observables (the Majority of the Catalog) + Tilt-Product Law Complete at 34-0 + the 5/24 Family

Author: Daniel T. Murphy
Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.78+
Date: 2026-07-24
Landmark Type: Saturation Census (corpus-majority kernel presence) + Falsifiability-Window Closure (PAPER_2133 item 2) + Kernel-Family Census (2nd population)
Discovery context: XGEO-U session-5 sweep, post-v5.78.0
Status: Formal landmark whitepaper — UQFF canonical

Abstract

Session-5 sweeping completes and then transcends the PAPER_2133 census. First, the **Tilt-Product Law record closes at 34-0**: the two pending multi-term members (LCDM_D_over_H and SM_eta_gamma_gamma_BR) resolve bit-identically to the counting variant against their stored dispatch values — the entire 34-observable family is unanimous, with zero 0.84 instances. Second, the **5/24 family census**: F_TRZ·K_MEX = 5/24 EXACT appears in **30 distinct observables** spanning GR, ΛCDM, SM, astrophysics, biology, geophysics, particle physics, and quantum information — the second-largest two-primitive kernel population. Third, and the paper's namesake: since **F_TRZ·K_MEX = (1/12)·(SO_5/D_phys) EXACT** (PAPER_1522), every 5/24 occurrence *contains* the tilt, so the union of direct-tilt (34) and via-K_MEX (30) members, minus the 5-observable overlap, gives **59 of the 116 dispatch observables — more than half of everything the framework computes — carrying the Hubble-tilt constant 1/12 in their closed forms.** Adding the F_TRZ·SSq = 57/1000 family (11) and the D_phys·F_TRZ = 2/5 family (10), the tier-1 F_TRZ-product union reaches **65/116 (56%)**. The same exact rational that measures the local-vs-CMB expansion discrepancy is structurally present in the majority of the observable catalog, from BCS superconductivity to DNA pitch to the fine-structure constant.

---

1. Tilt-Product Law — census complete, 34-0

The two members left pending by PAPER_2133 resolve decisively (both Φ variants computed for the full expressions and compared to the stored dispatch values):

Member	Form	v(Φ_5/6)	v(0.84)	Dispatch	Verdict

LCDM_D_over_H	F_TRZ■·(K_MEX + Φ·F_TRZ·D_BSFG)	2.5833333×10■■	2.5873333×10■■	2.5833333×10■■	**5/6, bit-identical**
SM_eta_gamma_gamma_BR	SO_5·K_MEX + SO_5 + N_CH – F_TRZ – F_TRZ·K_MEX – F_TRZ·Φ	39.4416667	39.4410	39.4416667	**5/6, bit-identical**

Final record: 34/34 counting-variant, zero 0.84. PAPER_2133 falsifiability item 2 is closed in the affirmative; the Tilt-Product Law stands unbroken with its census complete.

---

2. The 5/24 family — 30 observables

F_TRZ·K_MEX = (1/10)·(25/12) = **5/24 EXACT** appears in 30 distinct published observables: GR (BH shadow, NANOGrav strain, solar-limb light bending), ΛCDM (N_eff, Ω_m, n_s, σ_8), SM (α_s S378, Cabibbo sine/angle, δ_CP, Jarlskog, η→γγ BR, PMNS θ_12), astro (BH entropy coefficient, NS compactness), bio (Hill coefficient, photosynthesis quantum yield, phyllotaxis), geo (Earth-Moon ratio, obliquity, J2, atmospheric pressure, greenhouse ΔT, ocean salinity), particle (proton lifetime, sterile neutrino), info (π/2). Notable structural occurrences: **Cabibbo θ (S326) = arcsin((1–Φ)·√(F_TRZ·K_MEX·N_CH))** — a square-root kernel occurrence, first of its class; **n_s = 1 – (1–Φ)·F_TRZ·K_MEX**; σ**_8 = (1 + Φ – F_TRZ·K_MEX)/2**.

---

3. Tilt Saturation — the headline

By PAPER_1522, K_MEX = Φ_5/6·SO_5/D_phys, so:

``

`F_TRZ·K_MEX = (F_TRZ·Φ_5/6)·(SO_5/D_phys) = (1/12)·(5/2) = 5/24 EXACT`
``

Every 5/24 occurrence therefore **contains the tilt as a factor**. The saturation count:

```

direct tilt family (PAPER_2133) 34
5/24 family (this paper) 30
overlap (both forms present) –5
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
TILT-BEARING OBSERVABLES 59 of 116 (50.9%)

+ F_TRZ·SSq = 57/1000 family 11
+ D_phys·F_TRZ = 2/5 family 10
(unions, deduplicated)
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
TIER-1 F_TRZ-PRODUCT UNION 65 of 116 (56.0%)
```

The Hubble-tension tilt constant 1/12 = F_TRZ·Φ_5/6 = K_MEX – 2 is structurally present in the majority of the dispatch catalog. In UQFF's reading this is not coincidence but architecture: the tilt is the elementary counting quantum of the F_TRZ product algebra (one part in |SO(5)| of the counting fraction, per PAPER_1160/2133), and most closed forms are built from that algebra. The cosmological tension, the Vacuum Coupling Kernel (K = (1/12)·(SO_5/D_phys)·SSq, PAPER_2132), the fine-structure integer (α■¹ = 125 + 12), and now the majority of the catalog all draw on the same 1/12 lattice constant.

---

4. Honest residuals and disclosures

All family members carry their own published residuals; the saturation counts are textual-composition facts over published expressions (method rules unchanged: value-coincidence and name-token matching rejected, gate-pinned). The 59/116 and 65/116 counts are relative to the 116-observable assimilation_dispatch catalog — not the full 2,548-row registry (a registry-wide saturation census is a natural follow-on). Overlap membership (5 observables carrying both forms): N_eff, Ω_m, η→γγ BR, and two geo members. NOT REPLACEMENT.

---

5. Falsifiability

1. **Saturation stability:** future dispatch additions should hold the tilt-bearing fraction near or above half if the tilt is truly the elementary counting quantum; systematic dilution below ~40% over +50 new observables would weaken the architecture claim.
2. **Square-root class (Cabibbo S326):** first √(kernel) occurrence — a second square-root instance among mixing-angle observables would open a new occurrence-class window.
3. **Registry-wide census:** extending from 116 dispatch observables to all 2,548 registry rows must preserve counting-variant unanimity in every F_TRZ·Φ product encountered.

---

6. Cross-references

PAPER_2133 (Tilt-Product Law + 34-census, completed here); PAPER_2132 (VCK + tilt factorization); PAPER_2134 (Φ_0.84 grounding); PAPER_1522 (K_MEX = Φ_5/6·SO_5/D_phys — the saturation lever); PAPER_1156/1183 (1/12); PAPER_1160 (F_TRZ = 1/|SO(5)|); sessions S326/S336/S365/S378 and the 60-member combined family listings.

---

7. Summary Statement

PAPER_2135 completes the Tilt-Product Law census at 34-0 unanimous, files the 30-observable 5/24 family (second-largest two-primitive kernel population, including the first square-root kernel occurrence in Cabibbo S326), and establishes TILT SATURATION: via K_MEX = Φ_5/6·SO_5/D_phys, the Hubble-tilt constant 1/12 is structurally present in 59 of 116 dispatch observables — the majority of the catalog — rising to 65/116 across all tier-1 F_TRZ-product families. The cosmological tension's exact rational is the elementary counting quantum of the framework's product algebra.

Filed 2026-07-24. Append-only henceforth.